Detecting the Anomalous Activity of a Ship's Engine

Stakeholder Report

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1. Problem Overview

A shipping company's fleet is looking to get ahead of potential issues in its fleet by predicting potential anomalies within its fleet. The features critical to detect potential anomalies include **engine rpm, lubrication oil pressure, fuel pressure, coolant pressure, lubrication oil temperature and coolant temperature.** Issues with the engines could lead to engine malfunctions, potential safety hazards, and downtime causing delayed deliveries, resulting in the breakdown of a ship's overall functionality, consequently impacting the business, such as affecting revenue. Thus, by using various statistical and machine learning methods to predict anomalies and enable timely maintenance, the company aims to increase profit by reducing downtime, reducing safety risks for the crew, limiting fuel consumption, and increasing customer satisfaction through timely deliveries.

2. Overview of Approach

The methodology used was centered around traditional statistical models (Interquartile range - IQR) and unsupervised machine learning models (one class Support Vector Machine and Isolation Forest) (**Table 1**). This ensured a combination of univariate and multivariate techniques to achieve an efficient result. SVM and Isolation Forest are both unsupervised ML models and multivariate. The three anomaly detection techniques are outlined below:

Model	Type	Parameters included	Key Information
Interquartile range (IQR)	Statistical & Univariate	Individual features	IQR = Q3 - Q1 Lower bound = Q1 - 1.5 * IQR Upper bound = Q3 + 1.5 * IQR
One class Support Vector Machine (SVM)	Machine Learning & multivariate	Gamma, nu, kernel	Maps inputs points into a hyper-plane. A point is anomalous if it falls outside boundary
Isolation Forest	Machine Learning & multivariate	Contamination, n_estimators	-Uses random trees, views anomalies as data points

Table 1: Anomaly detection methods applied.

2.1. Anomaly Detection Workflow

The end-to-end process used for detecting the anomalies within the fleet is visualized below (**Figure 1**). This included preprocessing the raw data by finding missing and duplicate values, analyzing the descriptive statistics, and visualizing the data via boxplots and histograms.

Then proceeding to using anomaly detection via Interquartile range, one class SVM and Isolation Forest to derive anomalies that make up 1%-5% of the total data points.

PCA (Principal Component Analysis) provided data visualization for both machine learning models through dimensionality reduction.

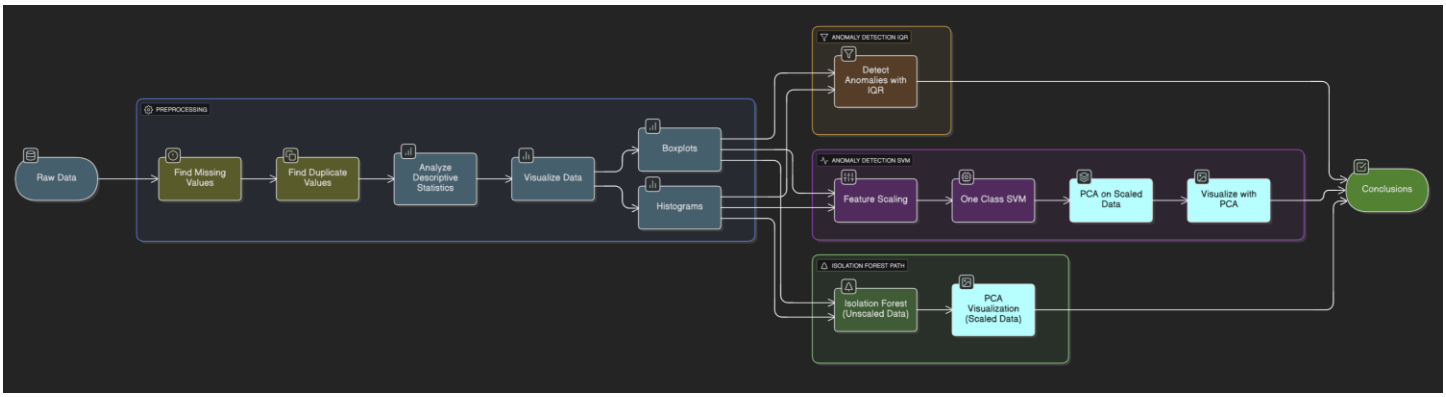

Figure 1: A high-level visualization of the anomaly detection pipeline.

2.2. Data and Features

There are 19,535 data points in the data set, each with 6 features, namely: *engine rpm, lubrication oil pressure, fuel pressure, coolant pressure, lubrication oil temperature and coolant temperature.* Based on the observations, the mean of the features surpassed their medians thus indicating skewness of each feature to the right (**Figure 2**).

Box plot visualization of the data reflected clear outliers largely focused above the top whisker of each feature. Notably, coolant temp reflected the least number of outliers within the features available. (**Figure 2**).

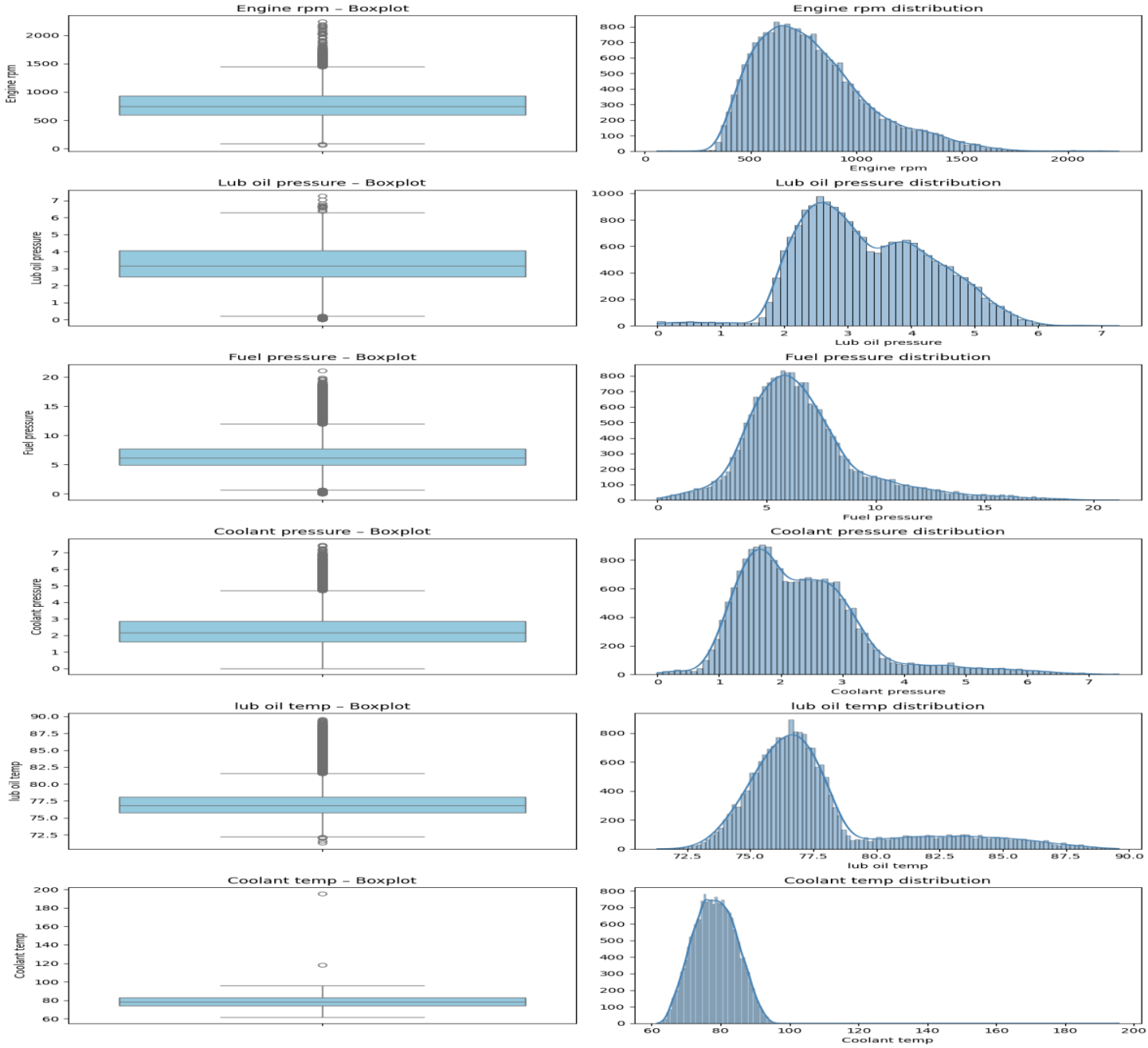

Figure 2: Visualization of ship fleet data per feature.

3. Interpretation of Anomaly Detection Results

Interquartile range (IQR): The IQR analysis conducted observed the percentage of ships with outliers in ***features > 0 (at least one feature) at 23.732%***, ***features > 1 at 2.160%, features > 2 at 0.056%.*** There were data points with outliers in more than 3 features. Anomalies were observed within ships with outliers in at least 2 features, which fall between the recommended range of 1%-5% at 2.160%. The total number of ships observed with these anomalies totaled 422.

One class Support Vector Machine (SVM): Different combinations of the nu and gamma were observed resulting in higher anomaly ranges as nu and gamma values rose (**Table 2**). Based on the observation, ***nu = 0.01 & gamma = 0.2*** indicated the best fit which aligned within the 1%-5% prescribed anomaly range. 202 ships with anomalies were observed at this nu & gamma, reflecting 1.034% of the total data.

It is important to note that while IQR observed 422 ships as having anomalies (at 2.16% of total data), SVM seems to be more efficient in narrowing down the observed data to provide the most accurate number of ships developing anomalies.

Gamma	nu	% of total data	No. of ships with anomalies
0.2	***0.01***	***1.034%***	***202***
0.3	0.03	3.015%	589
0.5	0.05	5.339%	1043

Table 2: Number of ships with anomalies with various Gamma and nu.

Isolation Forest: Using 100 isolation trees (n_estimators), observation from the isolation forest model asserted the ship data points with ***contamination 0.01-0.05*** all suitable as they all provided anomalies within the 1%-5% range (**Table 3**). Nevertheless, the most effective data points would be the anomaly closest to 1%. 1.003% of total data, totaling 196, contained ships with anomalies, very similar to the 202 ships observed with SVM model.

Contamination	% of total data	No. of ships with anomalies
0.01	***1.003%***	***196***
0.05	5.001%	589
0.1	10.003%	1954

Table 2: Number of ships with anomalies with various contamination values.

4. Evaluation of 2D PCA Plots

Principal Component Analysis (PCA) was used to visualize the SVM and Isolation Forest results whilst projecting the data onto a two-dimensional plane (**Figure 3**). The aim was to confirm the observation from both ML models that anomalies would form distinct clusters separate from normal data points. Most extreme anomalies appear on the periphery of the main data cloud, although the

most extreme outlier on both charts sits at the bottom right and would require further investigation and context.

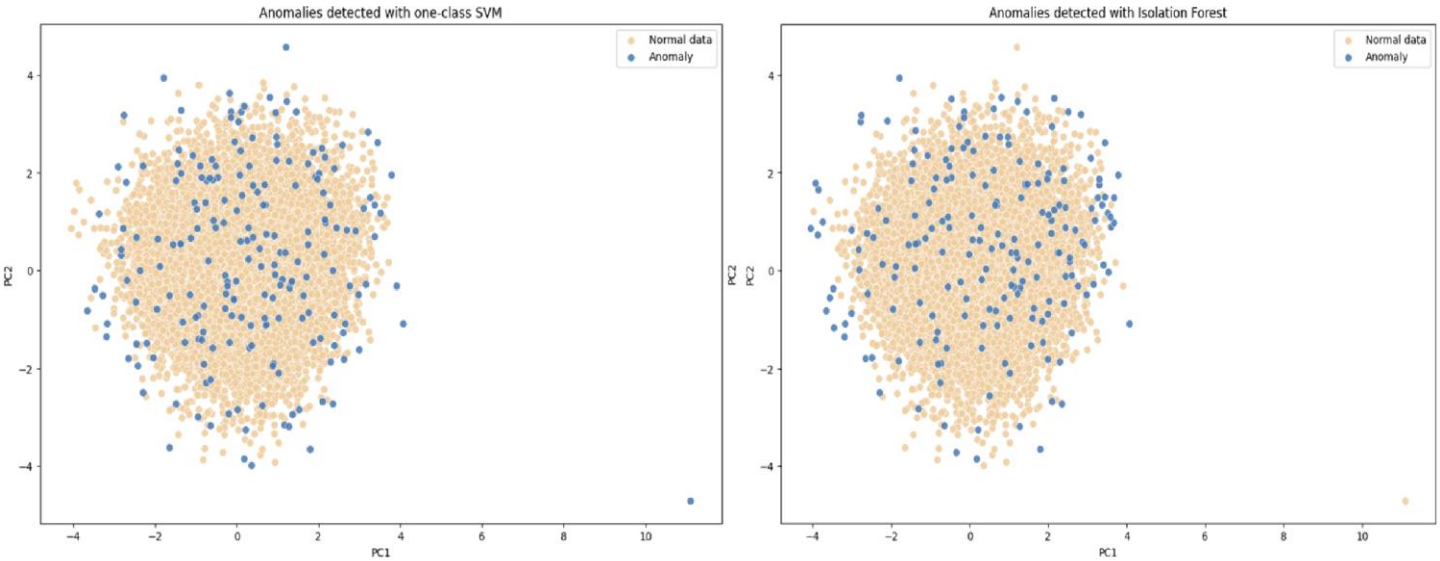

Figure 3: 2D PCA visualization of one class SVM and Isolation Forest.

5. Conclusions and Recommendations

- We observed a strong relationship in the number of ships identified as anomalies when employing SVM and Isolation Forest methods (both totalled at about 1% of the total data). These models reflect overlaps in the data observed as reflected in the visualised plots.

- The IQR model observes each feature individually and unlike the unsupervised ML models, requires the data to be labelled and organised. Its observations, even though falling within the 1%-5% range, should be used only when one class SVM and Isolation Forest have been exhausted, as these models have a more condensed anomaly selection.

- Overall, Isolation Forests would be the ideal choice as an anomaly detection model as it uses anomalies as data points, suitable for large data sets, computationally faster, uses a legion of isolation trees to achieve anomaly selection and most importantly, is not sensitive to feature-scaling and outliers like its one class SVM counterpart.

6. Next Steps

- Using the Isolation Forest anomaly detection model, there are 196 ships displaying anomalies, based on the data provided. The chief engineer and technical operations team should use this

data for further analysis including identifying the root cause of these anomalies and take corrective action.

- Findings should be duly conveyed to business stakeholders

- By using this robust anomaly detection system to predict timely maintenance, the business would be able to increase profit by reducing downtime, reducing safety risks for the crew, limit fuel consumption, and increase customer satisfaction through timely deliveries.

7. References

- Devabrat, M., 2022. Predictive Maintenance on Ship's Main Engine using AI. Available at: https://dx.doi.org/10.21227/g3za-v415 [Accessed 5 March 2024]